

Shruthi Kubatur, PhD

Clare Boothe Luce Assistant Professor of Computer Science | AI Systems | Multimodal Models

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Profile

Clare Boothe Luce Assistant Professor of Computer Science at the University of Texas at El Paso. My research program focuses on the reliability of autonomous AI agents: trajectory-level failure modeling, rare-event analysis of deployment risk, and causal root cause analysis of agent behavior and vision-language systems. This builds on a background in probabilistic modeling and rare-event simulation, and on eight years of post-PhD industry experience developing computer vision and multimodal models at Apple and other R&D organizations, where I led algorithms deployed in large-scale production systems.

Research Areas

Reliable and Trustworthy AI Systems

Multimodal and Vision-Language Models

Efficient Foundation Models and Edge AI

Computer Vision and Imaging Systems

Uncertainty Quantification and Rare-Event Modeling

Tech Stack: Python, PyTorch, JAX, MATLAB, L^AT_EX

Research Agenda

My long-term research program focuses on developing reliable AI systems that can operate safely and efficiently in real-world environments.

1. **Reliable AI Systems and Autonomy:** Developing methods to analyze and control the behavior of complex AI agents and multimodal systems operating in real environments.
2. **Efficient Multimodal Foundation Models:** Designing hardware-efficient architectures and optimization techniques for vision and multimodal models, particularly for edge and on-device inference.
3. **AI Systems for Real-World Perception:** Developing machine learning systems that integrate geometry, sensing, and multimodal reasoning for applications in imaging, augmented reality, and scientific domains.

Research and Industry Experience

Clare Boothe Luce Assistant Professor of Computer Science

Department of Computer Science, University of Texas at El Paso

El Paso, TX

Sept 2026 – Present

- Establishing a research group on reliability and failure analysis of autonomous AI agents
- Research areas: trajectory-level failure modeling, rare-event deployment risk, causal root cause analysis, and vision-language systems

Founder and Chief Scientist*Causyx AI.**Cupertino, CA**Jan 2026 – Present*

- Founded an AI systems startup focused on trajectory-level risk analysis for autonomous AI agents, analyzing execution paths to detect unsafe or unreliable behaviors
- Developing methods for causal analysis of agent trajectories to identify structural failure patterns in AI systems
- Designing tools for runtime monitoring and intervention in agentic AI workflows used in enterprise environments
- Leading product and research strategy for trustworthy AI infrastructure.
- Investigating structural failure patterns in autonomous AI systems to inform the design of more reliable agent architectures

Head of ML and Research*Happyverse AI.**San Francisco, CA**Oct 2025 – Dec 2025*

- Prototyped diffusion-based modeling approaches for imaging and computer vision tasks
- Defined ML strategy centered on multimodal and foundation modeling for early-stage AI products
- Built end-to-end modeling pipelines spanning data curation, training, and evaluation

Sr. Machine Learning Scientist, Video Engineering*Apple Inc.**Cupertino, CA**Oct 2020 – Oct 2025*

- Led work on model optimization (in collaboration with Apple China Vision Lab). Achieved 50% power and 66% inference-time optimizations. This increased model throughput and performance by designing hardware-efficient algorithms for Apple ML.
- Point-tracking algorithms for Apple proprietary applications
- Led development of multimodal vision-language models for spatial reasoning and 3D scene understanding in real-time perception systems
- Technical lead for machine learning algorithms improving camera imaging pipelines in Apple devices.

Sr. Software Engineer, Image Processing*Berkeley Lights**Emeryville, CA**Oct 2019 – Oct 2020*

- Led development of machine learning systems for biomedical and clinical imaging applications
- Developed machine learning and imaging algorithms for detection, segmentation, and classification of nanoscale cell images
- Implemented a novel live cell detection system for the antibody discovery pipeline, leading to an invention disclosure
- Devised an automatic bead decoding algorithm for BLI's novel OptoSeq™ Barcode BCR

Sr. Computer Scientist*Nikon Research Corporation of America**Belmont, CA**June 2017 – Oct 2019*

- Designed and trained deep learning models for image analysis, processing, and segmentation
- Implemented hardware-efficient machine learning inference pipelines
- Trained generative AI models (such as GANs) to generate data for proprietary applications
- Managed a synthetic data generation engineer

PhD Research Scholar*Purdue University**West Lafayette, IN**Jan 2014 – May 2017*

- Introduced a probabilistic framework for microstructure evolution in materials
- Developed a stochastic model for complex system behavior
- Adapted the Metropolis-Hastings algorithm to sample from importance-sampling densities to simulate rare events in material systems (collaborative research with Carnegie-Mellon University)

Algorithm Design Intern*ON Semiconductor**San Jose, CA**May 2016 – Aug 2016*

- Developed computer vision, HDR, and ISP algorithms for automotive image sensors
- Acquired real data for validation of algorithms for automatic driver assistance systems

Camera Imaging Intern*Aptina Imaging**San Jose, CA**May 2014 – Aug 2014*

- Created a Simulink library for simulation of the camera imaging pipeline
- Assisted in the design of an object detection framework

Graduate Assistant*University of Windsor**Windsor, ON, Canada**Sep 2010 – Apr 2012*

- Developed a neural network solution for unconstrained handwriting recognition
- Built the world's first Devanagari handwriting database with layered demographic texture

Select Research Projects

1. Rare-event simulation and probabilistic modeling for complex image structures (Purdue)
2. Efficient neural networks for real-time imaging systems (Apple)
3. Multimodal spatial reasoning models for AR systems (Apple)
4. Diffusion-based generative modeling for imaging tasks (Happyverse AI)
5. Structural failure analysis of autonomous AI agents (Causyx AI)

Teaching & Mentoring Experience

- Mentor to PhD students through Computing Research Association (CRA) programs
- Supervised engineers and early-career researchers in industry ML teams
- Lead TA for ECE 404 (Introduction to computer security) at Purdue University
- Primary instructor for ECE 308 (Systems, simulation, and control laboratory) at Purdue University
- TA for Math 161 (freshman calculus) at Purdue University
- TA for several ECE undergraduate laboratories at the University of Windsor, Canada

Funding

- Clare Boothe Luce Professorship, Henry Luce Foundation, 2026
- Faculty Startup Award, University of Texas at El Paso, 2026, \$478,168
- Principal Investigator, Corporate Exploratory Research Grant, Nikon HQ (Japan), 2018, \$100,000

Peer-Reviewed Publications

1. S. Kubatur, M. Comer, “Simulation of Rare Events in Images,” Computational Imaging at IS&T Electronic Imaging, February 2018.
2. S. Kubatur, M. Comer, “Rare Event Simulation for Markov Random Fields with Application to Grain Growth in Crystals,” IEEE International Conference on Image Processing (ICIP), September 2016.
3. S. Kubatur, M. Comer, “Simulation of abnormal grain growth in Polycrystalline Materials,” Computational Imaging at SPIE Electronic Imaging, 2016.
4. S. Kubatur, M. Comer, “Simulation of Grain Growth in Materials: A Signal Processing Perspective,” IEEE GlobalSIP, December 2015.
5. S. Kubatur, M. Ahmadi, M. Sid-Ahmed, “A Neural Network Approach to Online Devanagari Handwritten Character Recognition,” High-Performance Computing Symposium 2012.
6. S. Kubatur, S. Sreehari, R. Hegde, “An Image Processing Approach to Linguistic Translation,” Proceedings of the American Institute of Physics, 2011.

Patents and Invention Disclosures

1. Co-inventor on “Methods of Assaying a Biological Cell,” US Patent # 2023/0323433A1, March 2023.
2. Co-inventor on a US patent (provisional application) by Apple Inc. on spatial reasoning in LLMs, 2024.

Activities, Awards, Service

- Reviewer for several IEEE conferences and journals (including IEEE International Conference on Image Processing, IEEE ICASSP, IEEE Transactions on Computational Imaging)
- Expert reviewer for internal Apple machine learning conferences
- Session chair on PhD career paths (co-host with Prof. Jodi Tims / Northeastern University) at Computing Research Association (WP special edition), University of Minnesota, 2024
- Session chair on summer research internships (co-host with Prof. Cynthia Phillips / Sandia National Lab) at Computing Research Association (WP special edition), University of Minnesota, 2024
- 2024 Distinguished Alumna, BMS College of Engineering, Bengaluru, India
- Editorial member, Apple Video Engineering Newsletter
- Session chair (co-host with Dr. Kathryn McKinley / Google Research) at Computing Research Association (WP special edition), 2023
- Invited speaker at Computing Research Association (WP special edition), 2022, 2023
- Dean’s scholarship and International Graduate Excellence Scholarship, University of Windsor
- Chair of the “Modeling and Computation” session at IEEE GlobalSIP conference, 2015
- Volunteer at “Introduce a girl to engineering” program at Purdue University
- ECE Ambassador, Women in Engineering, Purdue University Chapter, 2015 – 2017

Education

PhD in Electrical Engineering
Purdue University, West Lafayette

May 2017

Dissertation: Stochastic Modeling and Rare-Event Simulation for Gibbs Distributions (Advisor: Prof. Mary L. Comer)

MS in Electrical Engineering

May 2012

University of Windsor, Canada

Master's thesis: Online Devanagari Handwritten Character Recognition (Advisors: Prof. Majid Ahmadi, Prof. Maher Sid-Ahmed)

BE in Telecommunications Engineering

July 2010

V.T.U., India

First Class with Distinction; Rank 2 in the whole cohort